

AN ISO 9001:2015 CERTIFIED INSTITUTE



CREATIVE

DESIGN & MULTIMEDIA INSTITUTE

Ch. 7

**Increment and Decrement Operator
Turnery Operator | Modulus Operator**

5. Increment and Decrement Operator in C :

Increment Operators are used to increase the value of the variable by one and Decrement Operators are used to decrease the value of the variable by one in C programs.

Both increment and decrement operator are used on a single operand or variable, so it is called as a unary operator. Unary operators are having higher priority than the other operators it means unary operators are executed before other operators.

Syntax :

`++` // increment operator

`--` // decrement operator

Note: Increment and decrement operators are can not apply on constant.

Example

`x= 4++;` // gives error, because 4 is constant

Type of Increment Operator :

- pre-increment
- post-increment

pre-increment (`++ variable`) :

In pre-increment first increment the value of variable and then used inside the expression (initialize into another variable).

Syntax :

`++ variable;`

Example pre-increment :

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int x,i;
    i=10;
    x=++i;
    printf("x: %d",x);
    printf("i: %d",i);
    getch();
}
```

Output :

x: 11

i: 11

In above program first increase the value of i and then used value of i into expression.

post-increment (variable ++)

In post-increment first value of variable is used in the expression (initialize into another variable) and then increment the value of variable.

Syntax :

variable ++;

Example post-increment

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int x,i;
    i=10;
    x=i++;
    printf("x: %d",x);
    printf("i: %d",i);
    getch();
}
```

Output

x: 10

i: 11

In above program first used the value of i into expression then increase value of i by 1.

Type of Decrement Operator:

- pre-decrement
- post-decrement

Pre-decrement (-- variable) :

In pre-decrement first decrement the value of variable and then used inside the expression (initialize into another variable).

Syntax :

-- variable;

Example pre-decrement

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int x,i;
    i=10;
    x=--i;
    printf("x: %d",x);
    printf("i: %d",i);
    getch();
}
```

Output

x: 9

i: 9

In above program first decrease the value of i and then value of i used in expression.

post-decrement (variable --):

In Post-decrement first value of variable is used in the expression (initialize into another variable) and then decrement the value of variable.

Syntax :
variable --;

Example post-decrement

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int x,i;
    i=10;
    x=i--;
    printf("x: %d",x);
    printf("i: %d",i);
    getch();
}
```

Output

x: 10
i: 9

In above program first used the value of x in expression then decrease value of i by 1.

Example of increment and decrement operator

Example:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int x,a,b,c;
    a = 2;
    b = 4;
    c = 5;
    x = a-- + b++ - ++c;
    printf("x: %d",x);
    getch();
}
```

Output
x: 0

6. Conditional or Ternary Operator (?:)

- C programming offers the conditional operator used in C language is (?) question mark and (:) colon.
- It is used for checking the condition and gives output according to the condition.
- If condition is true then it will execute the first value or it will execute the second value given in the syntax.

Syntax: **<Condition>? <True Value> :< False Value>**

Purpose:

If condition will be true then it will return the true value else it will return the false value.

Example:

```
a=10;
b=12;
c= (a>b)? A: b      // It will store value of B
printf("Value of C is %d",c);

sub1=23, sub2=40;

Printf ("Highest=%d", (sub1>sub2)? sub1:sub2)
```

Modulus operator in C :

The modulus operator is a symbol used in various programming languages. It is denoted by the percentage symbol (%). It is a modulus operator that is used in the arithmetic operator. It determines the remainder. In some cases, the remainder may be 0, it means the number is completely divisible by the divisor

Example :

```
Int a=13,b=2,c
```

```
C=a%b;
```

```
Printf("\nC=%d",c);
```

Output :

1

Assignment Work :

1. WAP to Find the Maximum/ Minimum number among the Three number Using Turnery Operator
2. WAP to check Odd / Even num of given Number